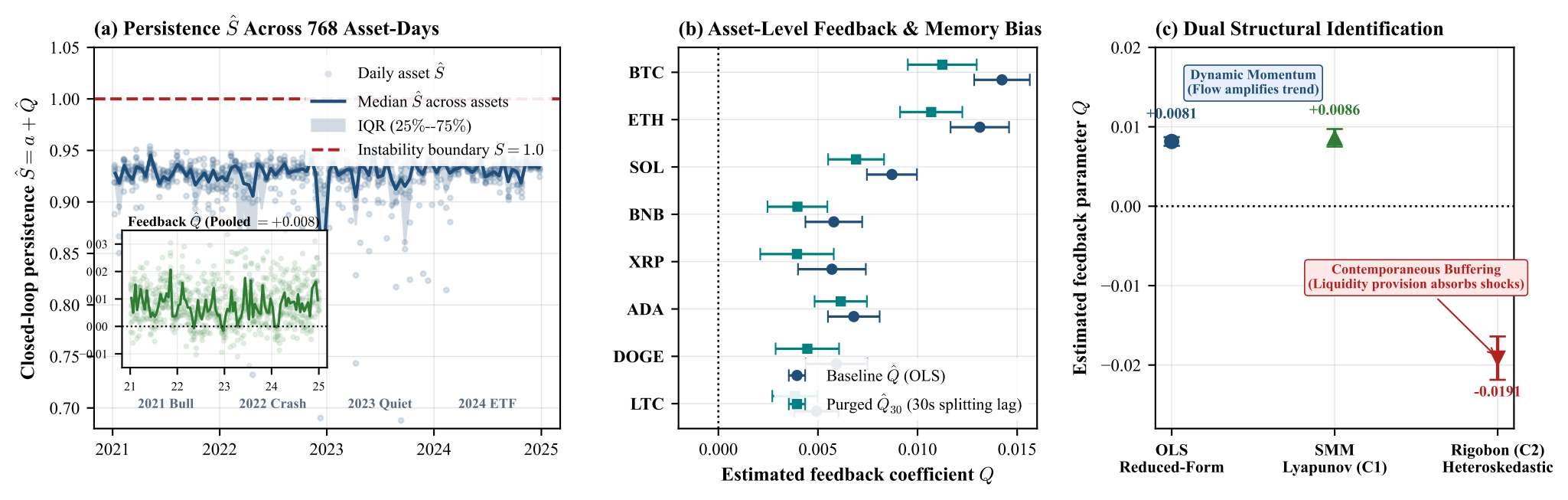

(a) Persistence $\hat{S}$ Across 768 Asset-Days
Closed-loop persistence $\hat{S} = a + \hat{Q}$
Daily asset $\hat{S}$
Median $\hat{S}$ across assets
IQR (25%--75%)
Instability boundary $S = 1.0$
Feedback $\hat{Q}$ (Pooled $= +0.008$)
2021 Bull
2022 Crash
2023 Quiet
2024 ETF
(b) Asset-Level Feedback & Memory Bias
BTC
ETH
SOL
BNB
XRP
ADA
DOGE
LTC
Baseline $\hat{Q}$ (OLS)
Purged $\hat{Q}_{30}$ (30s splitting lag)
Estimated feedback coefficient $Q$
(c) Dual Structural Identification
Estimated feedback parameter $Q$
Dynamic Momentum
(Flow amplifies trend)
+0.0081
+0.0086
Contemporaneous Buffering
(Liquidity provision absorbs shocks)
-0.0191
OLS
Reduced-Form
SMM
Lyapunov (C1)
Rigobon (C2)
Heteroskedastic